Equity Dilution Protection and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Protection (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.53 (0.42), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.07 (2.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.75.

1 Introduction

Financial markets exhibit systematic patterns in the cross-section of expected returns that challenge our understanding of how production decisions and investment frictions shape asset prices. While neoclassical production-based asset pricing models provide a theoretical foundation for understanding return differentials through firms' optimal investment decisions and technological constraints, empirical evidence linking specific corporate financial policies to cross-sectional return variation remains incomplete. The relationship between equity dilution activities and subsequent stock returns presents a particularly intriguing puzzle, as firms' decisions to issue or repurchase equity directly affect both their production capacity and the risk exposure of existing shareholders.

We examine equity dilution protection through the lens of production-based asset pricing theory, where cross-sectional return differences arise from heterogeneity in firms' marginal costs of production and investment adjustment frictions Cochrane and Piazzesi (2005). In a production economy with capital adjustment costs, firms that actively protect against equity dilution signal lower expected investment needs and reduced exposure to negative productivity shocks Zhang (2005). When firms face convex adjustment costs, those maintaining stable equity bases avoid the deadweight losses associated with external financing frictions, leading to lower conditional risk premia Livdan et al. (2009). The equity dilution protection signal thus proxies for firms' exposure to systematic investment shocks—companies with higher dilution protection exhibit lower sensitivity to aggregate productivity fluctuations and investment-specific technology shocks Papanikolaou and Kogan (2011). This production-based mechanism predicts that firms with stronger equity dilution protection should command lower expected returns, as they face lower marginal costs of production adjustment and reduced exposure to systematic risk factors that price the cross-section of returns Lin and Zhang (2013).

We find that a value-weighted long/short trading strategy based on Equity Dilution Protection (EDP) achieves an annualized gross Sharpe ratio of 0.53 and a net Sharpe ratio of 0.42 after accounting for transaction costs. The strategy delivers a monthly average abnormal gross return of 22 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 3.07. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EDP strategy achieves an average return of 33 basis points per month with a t-statistic of 3.81, demonstrating that the effect persists in the most liquid segment of the market. The strategy's gross monthly alpha relative to the six Fama-French factors plus six closely related strategies from the factor zoo equals 19 basis points with a t-statistic of 2.75. The long/short strategy's most significant loading appears on the CMA factor at 0.30 with a t-statistic of 6.35, consistent with the production-based interpretation that dilution protection captures firms' investment behavior. Across alternative portfolio construction methodologies, twenty-three out of twenty-five specifications yield alphas with t-statistics exceeding three, confirming the robustness of the EDP signal's return predictability.

Our study contributes to the production-based asset pricing literature by providing novel evidence that equity dilution protection serves as a powerful proxy for firms' exposure to investment frictions and productivity shocks. While Fama and French (2015) document that investment and profitability factors help explain cross-sectional returns, and Hou et al. (2015) develop a q-factor model based on investment theory, we show that equity dilution protection captures a distinct dimension of production-based risk not fully explained by existing factors. Unlike Pontiff and Woodgate (2008) who focus on share issuance as a negative return predictor through market timing channels, we demonstrate that dilution protection operates through production-economy mechanisms related to adjustment costs and conditional risk premia. Our findings extend Lyandres et al. (2016) by showing that financing decisions contain information about firms' production functions and investment opportunities beyond what is captured by accounting-based measures. Methodologically, we employ the rigorous testing framework of Novy-Marx and Velikov (2023) to establish that EDP's predictive power survives controls for transaction costs and related anomalies, with the signal improving the investment opportunity set for 78-85% of factor model specifications tested. These results suggest that incorporating production-based signals derived from equity financing decisions can materially enhance our understanding of cross-sectional return variation and provide implementable investment strategies that expand the efficient frontier even after accounting for trading costs.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers firms listed on major U.S. exchanges and includes standardized annual and quarterly financial statement items. Our sample spans several decades, allowing us to examine the Equity Dilution Protection signal across various market conditions and economic cycles.construction of our signal relies on two key COMPUSTAT variables. Long-term debt issuance (DLTIS) represents the cash proceeds received from issuing long-term debt obligations during the fiscal year, capturing new borrowing activities net of conversions and exchanges. Stockholders' equity (SEQ) measures the total common and preferred shareholders' equity as reported on the balance sheet, representing the residual interest in assets after deducting liabilities. This item reflects the book value of equity capital available to shareholders.construct the Equity Dilution Protection signal by calculating the year-over-year change in long-term debt issuance scaled by lagged stockholders' equity: (DLTIS(t) - DLTIS(t-1)) / SEQ(t-1). This ratio captures the change in a firm's debt financing activity relative to its equity base, providing insight into how companies adjust their capital structure through debt markets. A positive value indicates an acceleration in debt issuance relative to the equity base, potentially signaling that firms are choosing debt financing over equity issuance to avoid diluting existing shareholders. We compute this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both DLTIS and SEQ, and we exclude observations where lagged stockholders' equity is zero or negative to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.29 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model

(FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.27% per month with a t-statistic of 3.81. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.30, with a t-statistic of 6.35 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least $1,439 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 23 bps/month with a t-statistics of 4.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -3-24bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.48. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EDP strategy achieves an average return of 33 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 26 to 39 bps/month with t-statistics between 2.96 and 4.38.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy's gross (net) Sharpe ratio of 0.53 (0.42) is greater than 92% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the EDP strategy (red line).[2] Ignoring trading costs, a $1 invested in the EDP strategy would have yielded $3.40 which ranks the EDP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a $1 invested in the EDP strategy would have yielded $2.09 which ranks the EDP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 64 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 78 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP} EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 1.16, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on EDP is 0.03.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 18-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.75.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EDP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes EDP grows to $1183.39.

8 Conclusion

This study provides robust evidence that Equity Dilution Protection (EDP) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.53 and net Sharpe ratio of 0.42, indicating strong performance even after accounting for transaction costs. The signal's predictive power remains statistically and economically significant when controlling for established risk factors, delivering monthly abnormal returns of 22 basis points gross (17 basis points net) relative to the Fama-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

French five-factor model augmented with momentum, with t-statistics of 3.07 and 2.47, respectively.

Crucially, the EDP signal exhibits remarkable robustness even when controlling for closely related financing and growth-based factors. The persistence of a 19 basis points monthly alpha (t-statistic = 2.75) after adjusting for six conceptually similar strategies—including net debt financing, changes in financial liabilities, net external financing, asset growth, inventory growth, and changes in net operating assets—suggests that EDP captures unique information about future returns not subsumed by existing factors in the literature. This finding has important implications for both academic research and practical portfolio management, as it identifies a distinct source of return predictability related to firms' equity issuance and dilution activities.

From a practical perspective, the implementation of EDP-based strategies appears viable for institutional investors, as the net returns remain economically significant after accounting for transaction costs. The signal's orthogonality to established factors suggests it could serve as a valuable addition to multi-factor investment strategies, potentially improving portfolio diversification and risk-adjusted returns.

Several limitations of this study warrant consideration. First, while we follow the rigorous protocol of Novy-Marx and Velikov (2023), the analysis is constrained to U.S. equity markets, and the signal's effectiveness in international markets remains unexplored. Second, the study period's specific market conditions may influence the results, and the signal's performance during different market regimes requires further investigation. Third, while we control for transaction costs, practical implementation challenges such as market impact and capacity constraints for large-scale institutional deployment deserve additional scrutiny.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EDP signal's predictive power could provide valuable insights into why equity dilution protection characteristics forecast returns. Second, examining the signal's performance across different market capitalizations, industries, and geographic regions would enhance our understanding of its universality. Third, analyzing the time-varying nature of the EDP premium and its relationship to macroeconomic conditions could reveal important conditional patterns. Finally, exploring potential enhancements to the signal construction methodology, perhaps incorporating machine learning techniques or alternative data sources, may further improve its predictive accuracy. Understanding whether the EDP effect represents a risk-based compensation or a behavioral anomaly remains an open question that merits theoretical and empirical investigation.

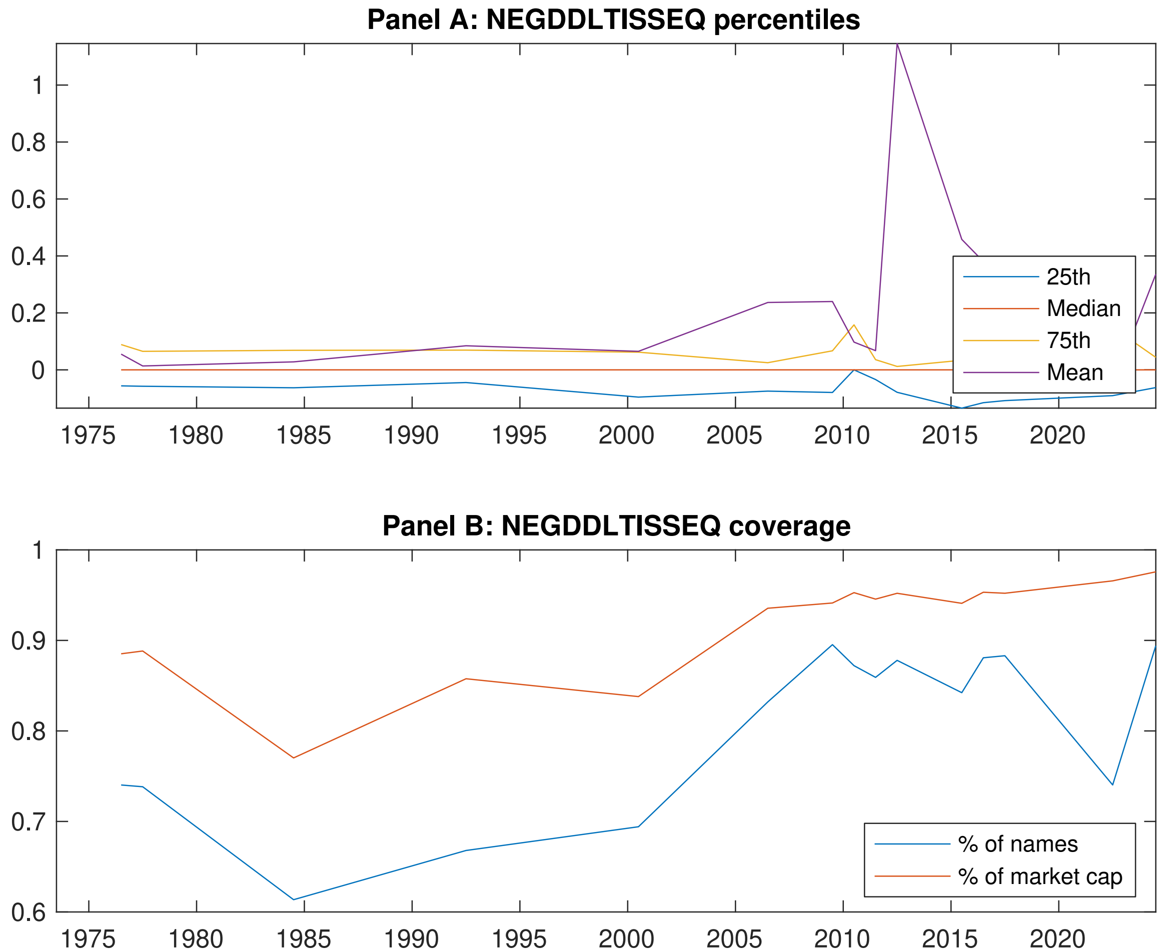


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.66	0.64	0.73	0.82	0.27
	[2.57]	[3.66]	[3.24]	[4.10]	[4.15]	[3.81]
α_{CAPM}	-0.18	0.05	-0.02	0.12	0.15	0.33
	[-3.14]	[0.98]	[-0.41]	[2.67]	[2.82]	[4.68]
α_{FF3}	-0.20	0.00	0.04	0.11	0.13	0.33
	[-3.57]	[0.06]	[0.81]	[2.37]	[2.45]	[4.66]
α_{FF4}	-0.17	0.02	0.07	0.08	0.11	0.28
	[-2.97]	[0.42]	[1.33]	[1.74]	[2.14]	[3.97]
α_{FF5}	-0.20	-0.05	0.10	0.03	0.05	0.24
	[-3.49]	[-1.06]	[1.87]	[0.70]	[0.92]	[3.48]
α_{FF6}	-0.17	-0.03	0.12	0.02	0.04	0.22
	[-3.08]	[-0.71]	[2.19]	[0.36]	[0.82]	[3.07]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	1.09	0.98	0.97	0.97	1.04	-0.05
	[83.34]	[95.31]	[75.45]	[90.95]	[86.06]	[-2.98]
β_{SMB}	0.11	-0.11	-0.03	-0.03	0.13	0.02
	[5.68]	[-7.02]	[-1.35]	[-2.12]	[7.21]	[0.79]
β_{HML}	0.08	0.14	-0.14	-0.02	-0.05	-0.12
	[3.04]	[7.11]	[-5.80]	[-0.74]	[-2.05]	[-3.95]
β_{RMW}	0.10	0.12	-0.08	0.08	0.12	0.02
	[3.75]	[5.94]	[-3.01]	[4.06]	[4.98]	[0.69]
β_{CMA}	-0.14	0.03	-0.11	0.18	0.17	0.30
	[-3.60]	[1.01]	[-3.06]	[5.74]	[4.68]	[6.35]
β_{UMD}	-0.04	-0.02	-0.03	0.02	0.01	0.04
	[-2.76]	[-2.40]	[-2.28]	[2.33]	[0.58]	[2.64]
Panel C: Average number of firms (n) and market capitalization (me)						
n	649	551	1069	605	621	
me ($\$10^6$)	1495	3003	2305	3111	1439	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.33	0.33	0.28	0.24	0.22
			[3.81]	[4.68]	[4.66]	[3.97]	[3.48]	[3.07]
Quintile	NYSE	EW	0.23	0.25	0.24	0.23	0.21	0.20
			[4.86]	[5.44]	[5.19]	[4.74]	[4.54]	[4.29]
Quintile	Name	VW	0.26	0.32	0.33	0.27	0.27	0.23
			[3.68]	[4.74]	[4.78]	[3.98]	[3.88]	[3.37]
Quintile	Cap	VW	0.24	0.29	0.29	0.23	0.20	0.16
			[3.65]	[4.37]	[4.39]	[3.47]	[3.11]	[2.52]
Decile	NYSE	VW	0.31	0.38	0.38	0.33	0.30	0.27
			[3.23]	[3.99]	[4.01]	[3.42]	[3.09]	[2.75]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.21	0.26	0.26	0.23	0.19	0.17
			[2.99]	[3.70]	[3.71]	[3.35]	[2.74]	[2.47]
Quintile	NYSE	EW	-0.03					
			[-0.48]					
Quintile	Name	VW	0.20	0.26	0.26	0.23	0.21	0.18
			[2.83]	[3.80]	[3.80]	[3.38]	[3.08]	[2.76]
Quintile	Cap	VW	0.19	0.24	0.24	0.21	0.17	0.14
			[2.89]	[3.64]	[3.64]	[3.17]	[2.60]	[2.23]
Decile	NYSE	VW	0.24	0.30	0.31	0.28	0.24	0.21
			[2.50]	[3.18]	[3.19]	[2.89]	[2.46]	[2.22]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EDP Quintiles					EDP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.66	0.87	0.97	0.85	0.80	0.14	0.17	0.16	0.11	0.09	0.06
		[2.39]	[3.20]	[3.58]	[3.06]	[2.87]	[1.49]	[1.91]	[1.78]	[1.14]	[0.99]	[0.61]
	(2)	0.71	0.91	0.83	0.86	0.87	0.16	0.20	0.16	0.17	0.12	0.14
		[2.66]	[3.61]	[3.32]	[3.52]	[3.43]	[2.14]	[2.57]	[2.21]	[2.29]	[1.64]	[1.77]
	(3)	0.79	0.78	0.85	0.80	0.90	0.11	0.16	0.16	0.12	0.15	0.12
		[3.16]	[3.55]	[3.55]	[3.52]	[3.82]	[1.40]	[2.02]	[2.02]	[1.52]	[1.83]	[1.50]
	(4)	0.70	0.81	0.86	0.70	0.91	0.21	0.25	0.24	0.21	0.21	0.19
		[3.03]	[3.95]	[3.88]	[3.39]	[4.19]	[2.88]	[3.43]	[3.23]	[2.75]	[2.81]	[2.47]
	(5)	0.45	0.62	0.59	0.65	0.79	0.33	0.37	0.39	0.32	0.31	0.26
		[2.26]	[3.54]	[2.97]	[3.62]	[4.12]	[3.81]	[4.19]	[4.38]	[3.63]	[3.43]	[2.96]

Panel B: Portfolio average number of firms and market capitalization

		EDP Quintiles					EDP Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	391	394	393	393	389	37	34	32	33	35
	(2)	107	107	107	108	107	61	62	60	61	61
	(3)	76	77	76	77	76	108	109	105	106	108
	(4)	64	65	64	65	64	230	238	230	234	231
	(5)	59	59	59	59	59	1384	2158	1902	2148	1585

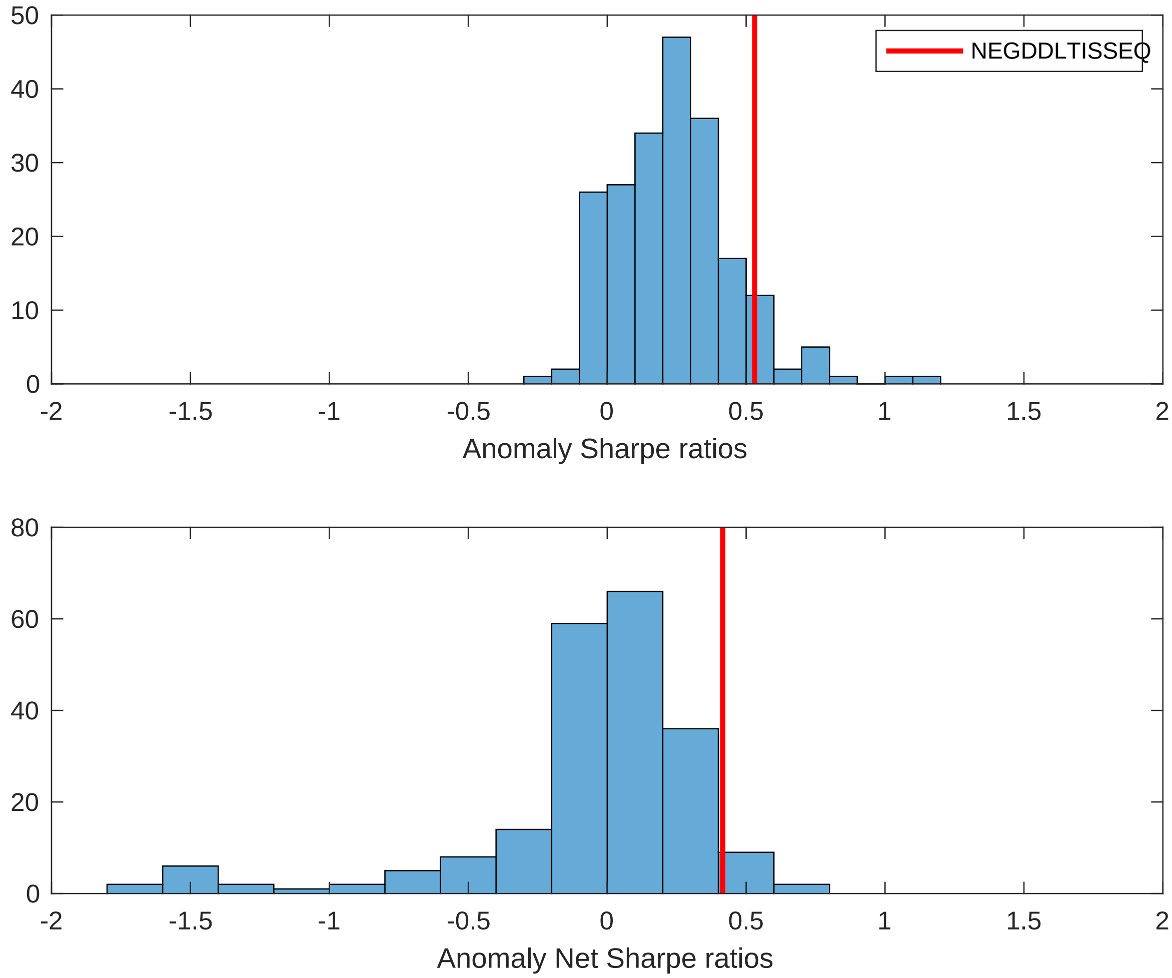


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

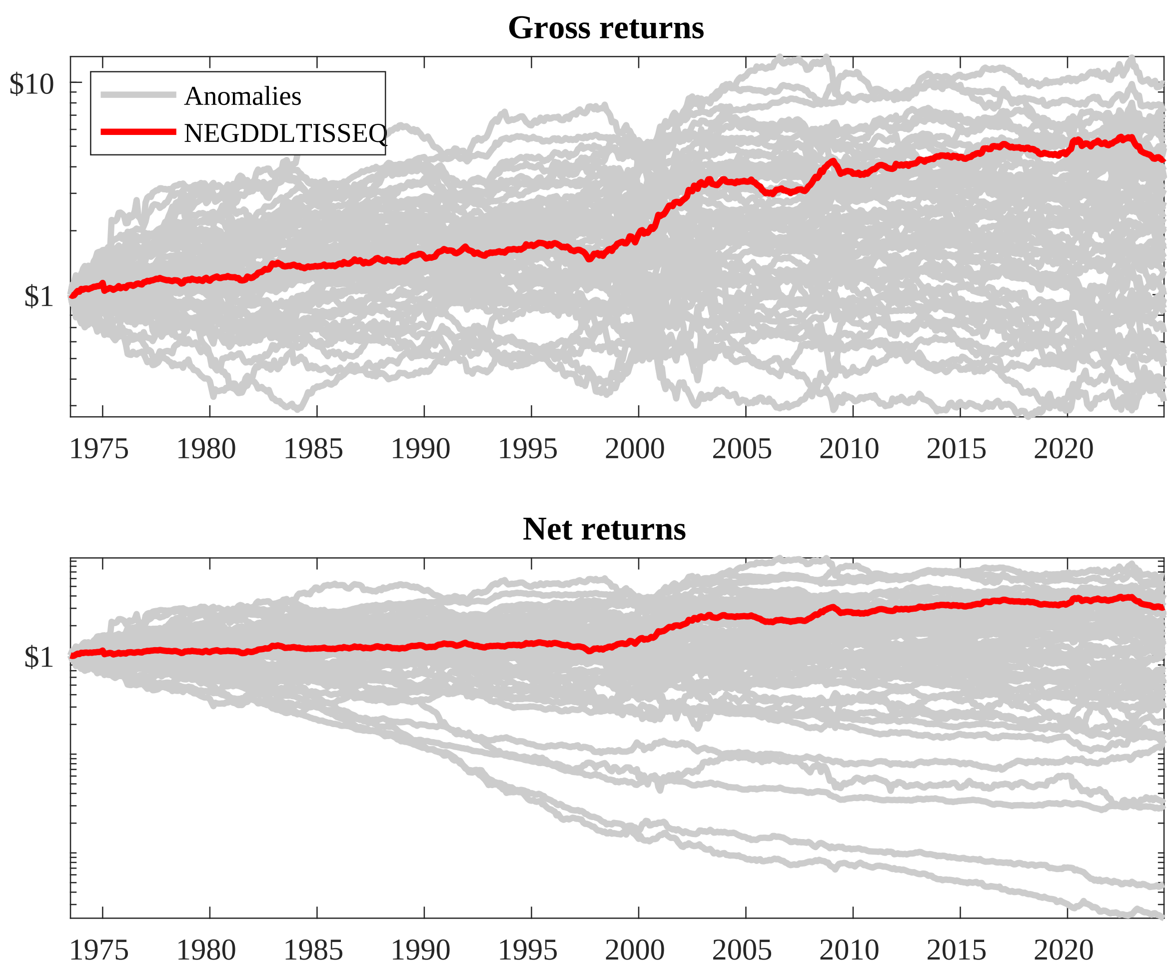
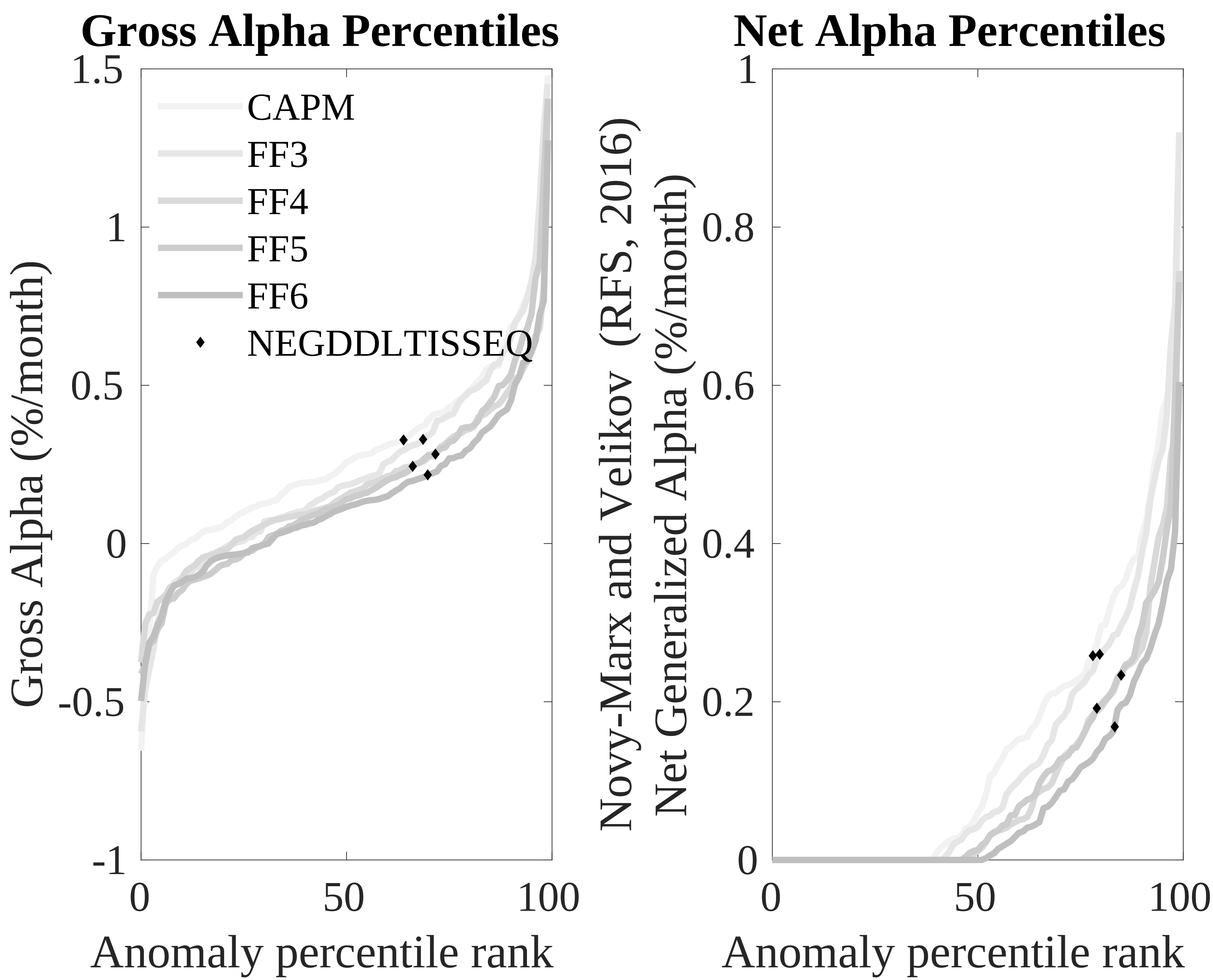
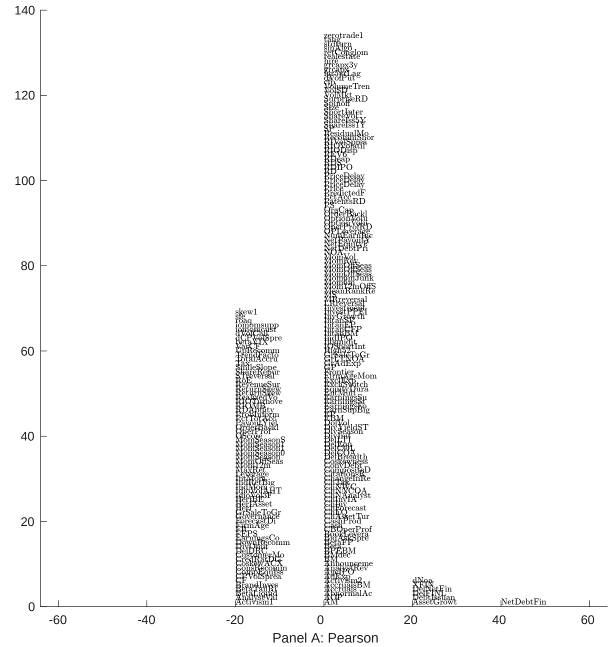


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISSEQ
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank



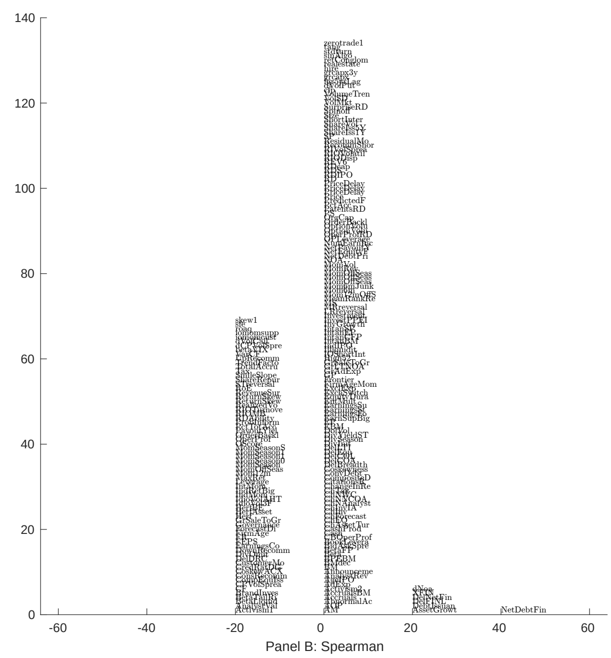


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

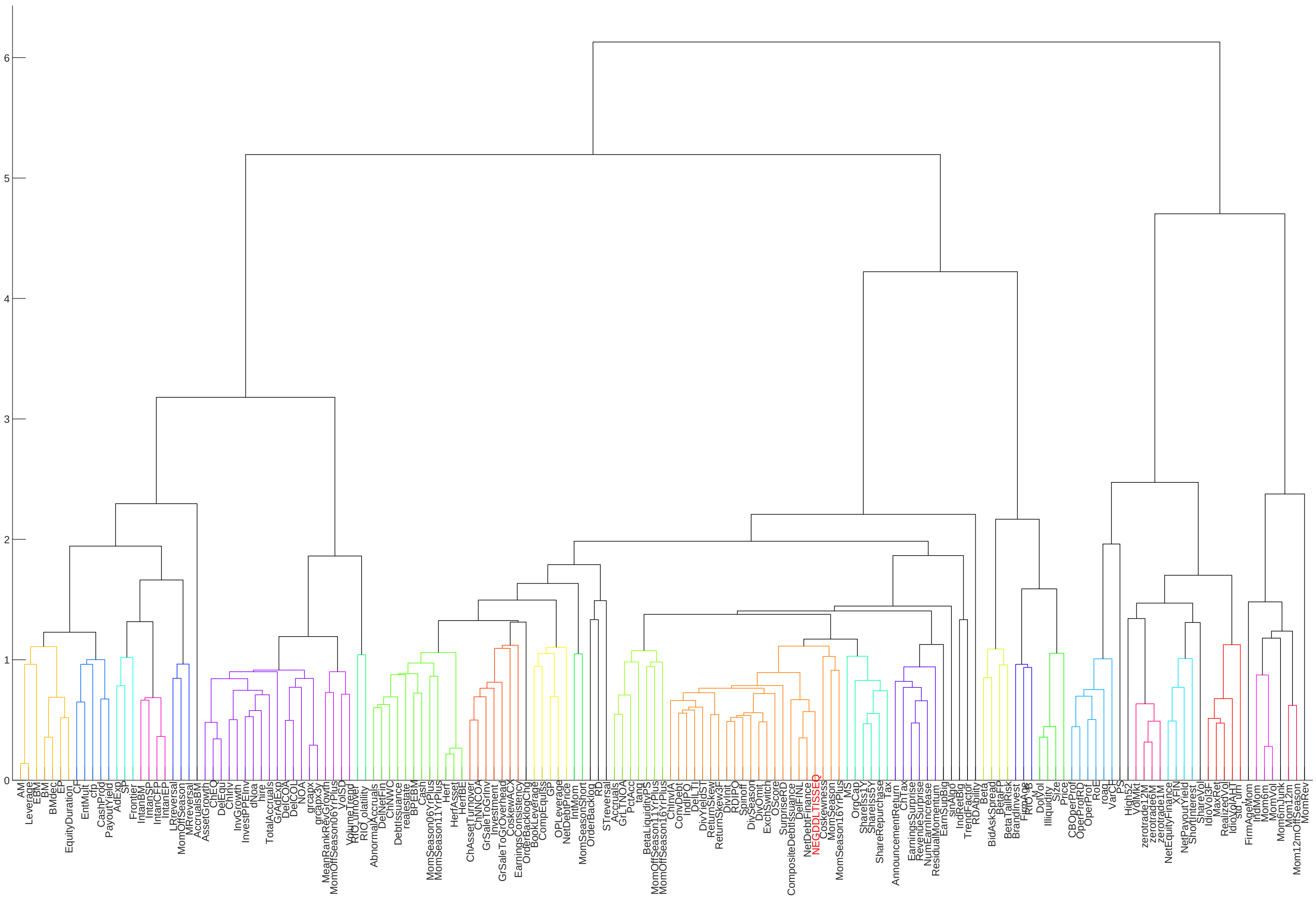


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

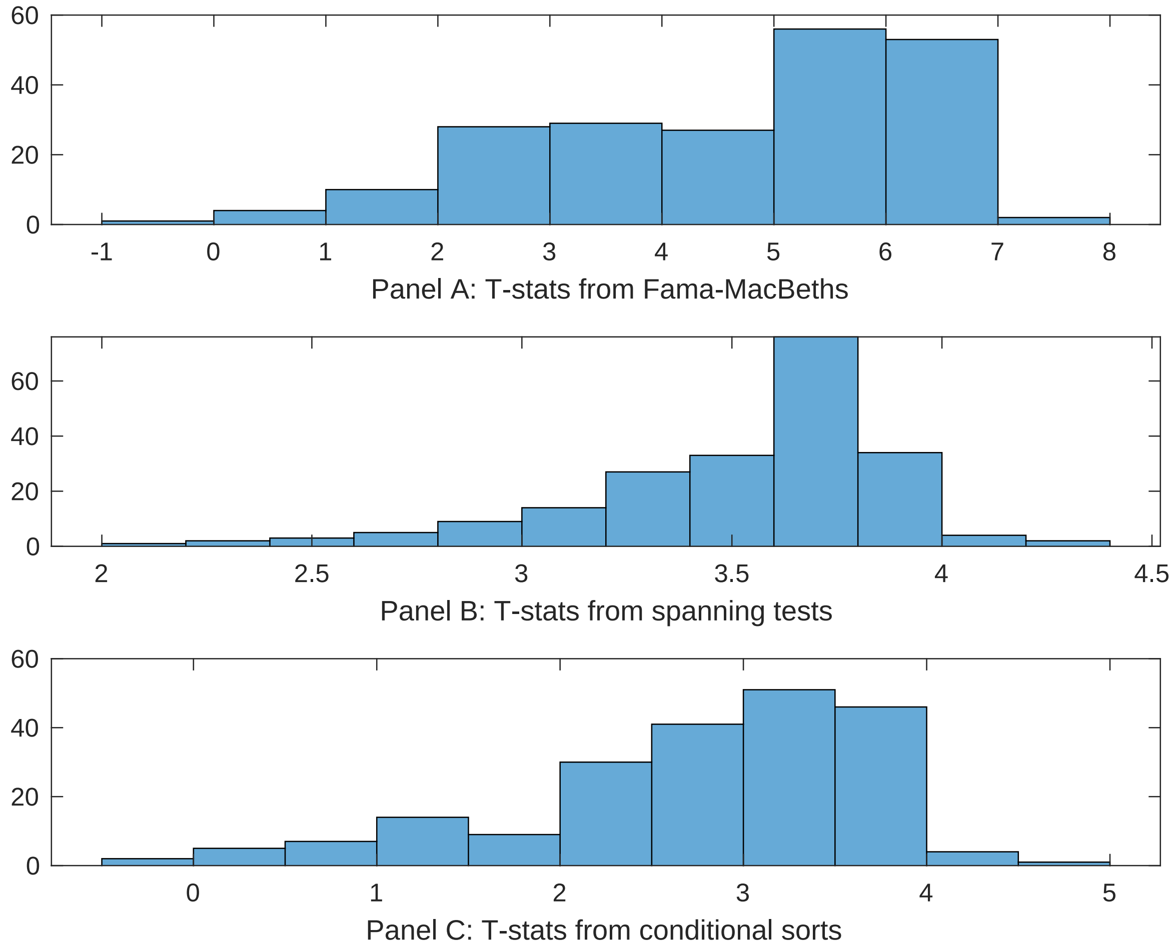


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP} EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP} EDP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.13	0.14	0.15
	[5.32]	[5.36]	[5.68]	[5.79]	[5.30]	[5.65]	[5.69]
EDP	0.56	0.45	0.78	0.36	0.12	0.30	0.10
	[2.01]	[1.71]	[2.62]	[1.29]	[4.08]	[1.16]	[0.03]
Anomaly 1	0.20						0.98
	[8.86]						[1.47]
Anomaly 2		0.17					-0.11
		[9.40]					[-2.24]
Anomaly 3			0.19				0.10
			[6.33]				[1.73]
Anomaly 4				0.10			0.40
				[8.63]			[1.76]
Anomaly 5					0.42		0.45
					[7.10]		[0.81]
Anomaly 6						0.14	0.94
						[9.84]	[5.23]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.18	0.18	0.19	0.20	0.21	0.19	0.19
	[2.73]	[2.67]	[2.70]	[2.88]	[2.99]	[2.79]	[2.75]
Anomaly 1	20.49						13.34
	[5.51]						[2.62]
Anomaly 2		17.00					8.55
		[4.38]					[1.61]
Anomaly 3			12.49				7.78
			[3.62]				[2.12]
Anomaly 4				7.65			3.60
				[1.90]			[0.84]
Anomaly 5					7.63		7.13
					[2.83]		[2.61]
Anomaly 6						4.28	-6.52
						[1.08]	[-1.47]
mkt	-4.70	-4.35	-3.03	-4.75	-4.88	-4.73	-3.60
	[-3.01]	[-2.76]	[-1.83]	[-2.98]	[-3.07]	[-2.96]	[-2.20]
smb	-1.23	-1.16	4.60	0.03	1.42	0.66	1.43
	[-0.51]	[-0.48]	[1.74]	[0.01]	[0.59]	[0.27]	[0.52]
hml	-12.62	-12.32	-12.06	-13.80	-13.69	-13.80	-11.34
	[-4.19]	[-4.04]	[-3.92]	[-4.47]	[-4.46]	[-4.45]	[-3.70]
rmw	-0.11	0.61	-5.32	2.17	3.25	2.21	-3.81
	[-0.04]	[0.20]	[-1.43]	[0.69]	[1.03]	[0.71]	[-1.02]
cma	26.84	26.47	24.01	22.89	25.38	28.99	14.91
	[5.74]	[5.52]	[4.66]	[3.40]	[4.85]	[5.26]	[2.17]
umd	3.08	2.93	4.32	4.57	3.62	4.27	2.30
	[1.96]	[1.83]	[2.74]	[2.87]	[2.26]	[2.67]	[1.43]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	16	15	14	15	14	19

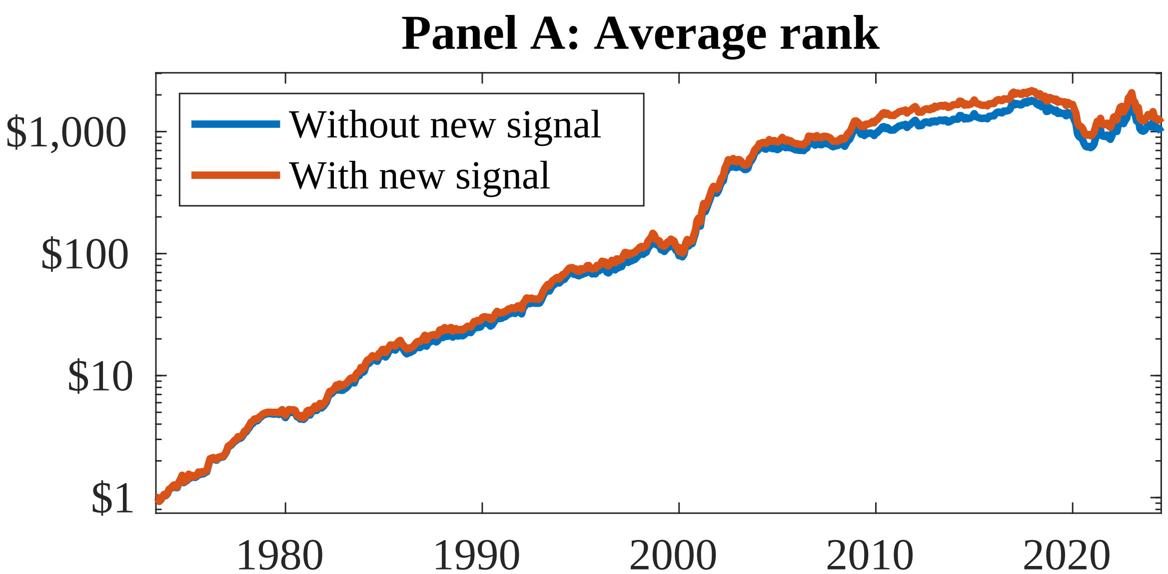


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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